

Validation Notes

Validation standard

Each AI-generated claim was checked against the statistical report, SQL analysis, and Power BI findings. Unsupported claims were excluded or labelled as hypotheses.

Validated findings

- 2,997 repeat customers versus 93,099 one-time customers; repeat customers represent 3.1188%.
- Mean order revenue is 138.6204 for one-time customers and 124.9113 for repeat customers.
- Average delivery delay is -11.8769 days.
- Average review score is 4.0864.
- Price and freight Pearson correlation is 0.4142; this indicates association, not causation.
- Potential revenue outliers represent 7.9575% under the IQR method.

Hypothesis validation

- H1: One-time and repeat customers have different revenue distributions. Mann–Whitney U was performed, but the numerical p-value/statistic is not reported; therefore significance is not claimed.
- H2: Delivery performance is associated with customer satisfaction. The report provides separate delivery and review statistics but no formal test between them; therefore causality is not claimed.
- H3: Revenue is concentrated in specific regions. Regional revenue analysis supports investigation, but the statistical report does not provide a formal concentration test; regional conclusions should therefore use SQL/Power BI evidence.

AI quality check

AI was not used to invent missing statistical results. Where the source did not provide a statistic or p-value, the limitation was recorded explicitly.